

Where the money comes from decides what breaks

Fraud, paywalls and the structure of Indian job platforms

A review-mining study of 421,387 Google Play reviews across seven apps · August 2026

1. What this is

I pulled every Google Play review I could get for seven Indian job apps, classified the complaints in the negative ones, and looked at whether the mix of complaints changed when each app shipped a new version.

The headline result was not the one I expected. I went in looking for release-driven regressions. What I found instead was that release history barely matters, and that the thing separating these platforms from each other is how they make money.

The apps that don't charge jobseekers carry roughly ten to thirteen times more employer-fraud complaints than pricing complaints. The one app that does charge sits at a ratio of about one to one. That gap has held steady for four years and 171 releases.

This document covers how the analysis was built, what I checked before trusting it, and where it falls down.

Apps in scope

App	Side	Play ratings	Store rating	Reviews pulled
apna	Jobseeker	749,256	4.70	50,000
Naukri	Jobseeker	2,008,047	4.62	50,000
WorkIndia	Jobseeker	351,400	4.28	50,000
foundit (Monster)	Jobseeker	86,824	4.62	15,136
Indeed	Jobseeker	13,499,317	4.47	250,000
WorkIndia Recruiter	Recruiter	12,842	3.53	5,759
Naukri Recruiter	Recruiter	2,986	4.28	492

Two of these are the same company on opposite sides of its own marketplace. That turned out to matter more than I expected when I added them.

2. Why job platforms

Quick commerce is the default subject for this kind of analysis and it is thoroughly picked over. Job platforms are less crowded, and they have a property that makes them more interesting: two sides with opposing interests. A jobseeker and a recruiter using the same product complain about completely different things, and comparing those complaints tells you something about the marketplace that neither side alone can.

Reviews are also the only honest public record of what breaks in these products. Nobody publishes their support ticket volumes. But 750,000 people have left public reviews of apna, and a decent share of the angry ones say precisely what went wrong.

3. Getting the data

I scraped with google-play-scraper, sorted newest first, filtering to English and India. Raw JSON went to disk before anything touched it, so the analysis could be re-run without re-scraping — which mattered, because I re-ran it a lot.

421,387 reviews came back. Then most of them had to go.

Stage	Reviews	What was removed
Raw scrape	421,387	—
6+ words	139,622	"good", "nice", emoji-only
Deduplicated	137,419	Repeated identical text
1–2 star only	29,988	Positive and neutral reviews

The word filter does more work than it sounds like. Over half the corpus is a single word — "good" appears 61,103 times on its own. Those reviews cannot be classified by any method, human or machine, and feeding them to a classifier just produces confident nonsense.

Restricting to one and two stars was a deliberate trade. The unit of analysis is complaints, so five-star reviews contribute nothing to the mix. But it costs the ability to say complaints went up in absolute terms — only that the mix shifted. I handled that by computing negative-review rate separately alongside complaint mix, so severity and composition are measured independently.

A problem I nearly missed

Different apps hit the scrape ceiling at different historical depths. Indeed's first 50,000 reviews only reached back five months, because it gets so many. apna's reached back eighteen. Comparing those two directly would have meant comparing 2026 against 2025 and calling the difference a platform difference.

Fix was to scrape Indeed deeper until it reached May 2024, then restrict all cross-app work to a common window of February 2025 to August 2026.

4. Building the taxonomy

I read 310 reviews by hand before writing any classification logic, and built the categories from what was actually there rather than from what I assumed would be there. This was the least automatable part of the project and probably the most important.

It also took three passes, because my first attempt was wrong in an instructive way. My initial list had separate codes for spam calls, privacy violations, and being asked for money. Reading the reviews, those three kept collapsing into each other — a fraudulent employer demanding a registration fee is also harvesting your number and also calling you repeatedly. Rather than force artificial splits I merged them, and fake_scam_posting went from 45 to 62 of my 310 labels.

The distinction I did keep, and which turned out to carry the whole analysis:

- `fake_scam_posting` — a third-party employer defrauded the user. Registration fees, fake vacancies, bait-and-switch, MLM.
- `credits_pricing` — the platform itself charged the user. Premium tiers, credits, paid resume services.

Both look like "they took my money" in the review text. They are entirely different product problems with entirely different fixes, and conflating them would have inverted the finding.

Final taxonomy: eleven categories plus other, with other running at 12.6% of the golden set.

5. The classifier, and how wrong it is

I had budgeted for an LLM tagger and could not fund it, so I built a rule-based classifier instead: about forty weighted regex patterns, deterministic, auditable, free to run. Every decision it makes can be traced to a specific rule, which is a genuine advantage over a model that cannot explain itself.

It is also not very good, and I would rather report that plainly than bury it.

Scoring

I split the 310 hand-labels into 201 for tuning and 109 held out. The held-out set was never used to write or adjust a single rule.

Measure	Value
Held-out accuracy	53.2%
Weighted precision	0.78
Weighted recall	0.53
Tuning-set accuracy	56.2%

The shape of that matters more than the headline. Precision runs 0.88 to 1.00 for account access, pricing, irrelevant matches and recruiter non-response. Recall is what is poor. The classifier does not guess wrong — it declines to guess. Roughly half of substantive negative reviews come back unclassified.

For share-of-complaints analysis that is survivable, provided the abstention is even. So I tested whether it was.

Full disclosure on the numbers

First evaluation came in at 49.5%. Inspecting the disagreements, a large share were cases where my own labelling had been inconsistent — I had put "this app often posts expired jobs" under recruiter posting tools, which is not what that category meant. I re-adjudicated 76 conflicting cases and re-scored, giving 53.2%.

That is a legitimate process, but it has to be stated. Reporting only the final number without saying the answer key was revised after seeing the predictions would be misleading.

6. Three checks that changed the analysis

Most of the work in this project was not producing findings. It was trying to break them.

Does the release-date inference actually work?

Play reviews carry the app version they were written against, but not the release date. I inferred each version's release as the earliest review carrying that tag. That assumption needed testing, and apna handed me a way to test it: apna uses date-encoded version strings like 2026.24.06.

Across 32 versions, inferred release dates landed a median of 6 days after the encoded date, with a 90th percentile of 10 days. Four apparent outliers at 34 to 37 days all trace to apna switching its version format in mid-2025, not to the method failing.

Consequence: a fourteen-day comparison window is too tight against a six-day lag. I widened everything to thirty days.

Does the classifier treat all apps equally?

No. Abstention was stable over time — 49% to 55% across eight quarters with no trend, which means within-app comparisons over time are sound. But across apps it varied by 29 points.

App	Abstention rate	After length standardisation
WorkIndia Recruiter	31.4%	35.0%
foundit	42.0%	39.5%
WorkIndia	42.8%	42.3%
apna	43.0%	42.8%
Naukri	45.4%	46.3%
Indeed	60.6%	61.4%

My first theory was review length — shorter reviews have fewer keywords to match, and abstention runs 66% on six-to-ten word reviews against 32% on forty-plus. But standardising every app to the same length mix barely moved the spread, from 29.2 points to 26.4.

The cause is Indeed specifically. It is the one global product in the set, and my taxonomy was built by reading mostly India-platform reviews. Its users describe a different product surface in different words, and my rules never learned them.

Consequence: Indeed is excluded from all cross-app comparison. It appears only in within-app version analysis, where the instrument only has to be consistent with itself. Losing my largest dataset was expensive, but publishing a cross-app chart where one bar is an artifact of my own vocabulary would have been worse.

Did anything actually change at a release?

171 versions across four apps had at least 40 negative reviews in both a thirty-day pre-window and a thirty-day post-window. Testing eight categories per version gives 1,368 comparisons.

Two exceeded two standard deviations. That is fewer than chance alone would produce.

The largest apparent shifts, before testing, were all the same category alternating sign: minus 14.5, minus 12.4, plus 12.4, plus 12.1, minus 11.3. That is one number oscillating around a stable mean with roughly a hundred reviews per window, not a sequence of product events. Picking the top twenty-four out of 1,368 comparisons and calling them findings is the standard way this kind of project generates fiction.

7. Findings

The market splits on fraud, and it splits on price

Share of substantive negative reviews:

App	Employer fraud	Platform pricing	Ratio
WorkIndia	42.1%	3.1%	13.4
apna	37.0%	3.5%	10.7
foundit	17.6%	3.3%	5.3
Naukri	15.3%	16.5%	0.9

The blue-collar and mass-market platforms run two and a half to three times Naukri's fraud share. Against a per-version noise floor of 3.9 to 6.2 points, a 24-point gap is four to six standard deviations. It holds quarter by quarter from mid-2024 through to August 2026.

Naukri is the only app in the set where pricing complaints outnumber fraud complaints. It is also the only one that charges jobseekers.

No release has moved it

WorkIndia's fraud-complaint share has oscillated around 42% across 51 releases spanning four years. Whatever the team has shipped, and they have shipped a great deal, none of it has durably changed what users complain about.

This reads as a negative result and is actually the more useful half of the finding. It means the problem is not in the backlog. It is in the business model.

Both sides of the same marketplace distrust the other

WorkIndia ships separate seeker and recruiter apps, which allows a clean comparison that does not depend on inferring who wrote what.

	Employer fraud	Platform pricing	Store rating
Seeker app	42.1%	3.1%	4.28
Recruiter app	28.0%	22.7%	3.53

Recruiters complain about pricing at seven times the seeker rate, and still report fraud at 28%. WorkIndia does monetise — it charges the recruiter side — and it has not bought trust in either direction. The paying side of its marketplace rates it three quarters of a star below the free side.

8. Sizing the top finding

Taking WorkIndia, the worst case. 18.2% of all scraped WorkIndia reviews describe employer fraud. Applied to 351,400 lifetime ratings, that implies roughly 64,000 users have publicly reported being defrauded.

Turning that into affected users requires an assumption, and I would rather show it than hide it inside a single confident number. Defrauded users are far more likely to review than average users, so the usual 1% consumer-app review rate is the wrong anchor.

If this share of defrauded users reviews	Implied affected users
20%	~320,000
10%	~640,000
5%	~1,280,000

The range is wide because the assumption is genuinely uncertain, and no amount of work on this dataset would narrow it. A number with a visible assumption is honest. The same number without one is invented.

9. Why this hasn't been fixed

Not incompetence, and not an unprioritised backlog. Policing fraud on a two-sided marketplace costs supply, and supply is the constrained side.

Naukri's paywall functions as a screening mechanism whether or not it was designed as one. Charging money — from either side — filters out actors for whom the fraud is only profitable at zero marginal cost. Naukri pays for this in pricing complaints, the highest in the set at 16.5%, and gets a fraud share of 15.3% in return.

apna and WorkIndia removed that friction deliberately, because it is the only way to reach the mass market. Charging jobseekers earning fifteen thousand rupees a month is not commercially viable, and charging small recruiters suppresses exactly the listings that make the platform useful in the first place. They inherited the fraud that friction was suppressing.

WorkIndia's two-sided data shows the trade is live rather than theoretical. It does charge recruiters, generating 22.7% pricing complaints, and still runs 28% fraud on that side. Partial monetisation bought partial screening at full complaint cost.

The uncomfortable part: neither position is obviously wrong. Enforcement that removes fraudulent listings also removes listings. On a platform where a user's alternative is no job at all, an aggressively policed empty marketplace may serve them worse than a populated risky one. That is the actual decision in front of these teams, and it is not solved by better machine learning.

10. What I would recommend

Verified-employer tier, not blanket enforcement. A visible verification badge gated on business registration and a returns-to-listing check. Unverified listings stay visible but carry a flag. This keeps

supply on the platform while giving seekers a signal to act on, and turns verification into something the recruiter side can be sold rather than a pure cost centre.

Metric: fraud-complaint share among reviews referencing verified-tier listings against baseline, measured at 90 days. Target: halve fraud share inside the verified tier without dropping total listing volume more than 10%.

What I would not recommend is aggressive delisting. WorkIndia's four-year flat line suggests incremental enforcement has already been tried and has not moved the number. The constraint is structural, so the intervention has to change the structure rather than push harder on the same lever.

11. Limitations

- App-store reviews are negativity-biased and self-selected. The rating distribution is bimodal — 79% five star, 12% one star. All shares are within the complaint population, not the user population.
- Android only. No iOS data at all. Probably minor for the blue-collar platforms, possibly not for Naukri and foundit.
- In-app review gating shapes who reviews. Apps that prompt after a successful action inflate positive volume, which distorts rate measures more than mix measures.
- Filtering to English suppresses some Hindi and Hinglish reviews. Latin-script Hinglish is captured; Devanagari largely is not. Since apna and WorkIndia skew blue-collar, this may bias their samples toward more English-fluent users.
- Classifier recall is 0.53. Roughly half of substantive negative reviews go unclassified. Precision is high, so what is counted is counted correctly, but every category is undercounted.
- Indeed is excluded from cross-app comparison because its abstention rate is 15 points above the next-worst app in every length bucket.
- Naukri Recruiter has only 296 reviews and is used qualitatively. It supports no rate or version analysis.
- apna's cross-app figure rests on 6 versions with sufficient data, against WorkIndia's 51.
- Severity scores are rule-assigned rather than measured, so they are excluded from every finding.

12. What I'd do differently

The classifier is the weak point and I know exactly how to fix it. An LLM tagger on the same 30,000 reviews would likely land above 80% agreement against the same golden set. The infrastructure is already written; it needs API credit and about two hours. Reporting both — rules at 53%, model at 80-something, findings holding under both — would turn the weakest section into the strongest.

I would also build the taxonomy from a sample stratified across all seven apps rather than weighted toward the Indian platforms. That single choice is what cost me Indeed, and I did not see it coming until the bias check surfaced it.

And I would extend tagging to three-star reviews. They are the most analytically useful ones — "useful app but too many spam calls" is a specific, credible, actionable complaint — and I cut them to save cost before I understood what I was giving up.